MIL-C-60505A(MU)
18 June 1970
SUPERSEDING
MIL-C-60505(MU)
30 September 1966

MILITARY SPECIFICATION

CAVITY ASSEMBLY: 11727549

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE EXERCISED IN OPERATING AND TESTING THE CAVITY ASSEMBLY: 11727549. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL, PAINT, ETC. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE CAVITY ASSEMBLY SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 This specification covers a solid state, mechanically Q-switched laser cavity which generates a single laser pulse at a wave length of 6943 Angstroms and contains an internal temperature sensing network. The cavity is a modular component of the Range Finder, LASER, AN/GVS-3 (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

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DRAWINGS

U.S. Army Frankford Arsenal

F8565556	Pressure Tester - Two Station
F11727549	Cavity Assembly: 11727549
F10540511	Housing
C10540690	Receptacle
C10540981	Plug
B10541011	Terminal

PACKAGING DATA SHEET

11727549	Packaging of Cavity Assembly

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The cavity assembly shall be manufactured in accordance with Drawing F11727549 and drawings pertaining thereto.

3.3 General specification.- The cavity assembly shall meet the following requirements, where applicable, of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Electrical and electronic assembly
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.3.1 Electronic workmanship.- In addition to meeting the workmanship requirements of MIL-F-13926, the cavity assembly shall meet the electronic workmanship requirements of 3.3.2 through 3.3.10.

3.3.2 Installation of hardware.- Installation of hardware on temperature sensing assembly shall be performed as specified on the drawing. The assembly shall not contain any of the following defects.

a. Cracks or deformation of the board adjacent to installed hardware.
b. Bent terminals or mechanical stand-offs.
c. Loose or improperly installed hardware.
d. Damage to the roll-over flange of the swagged hardware such that two or more full length cracks are located within a 45 degree angle.
e. Hardware from which the plating has been removed or peeled.

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3.3.3 Spacing of parts.- The parts of the temperature sensing assembly shall be spaced and mounted as specified on the applicable drawings.

3.3.4 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that any reference symbols on the board are not obscured, and that the proper identification markings of each part are visible.

3.3.5 Bend radius.- The minimum radius for all formed leads shall be one lead diameter. The maximum shall be two diameters or 0.062 inch, whichever is smaller.

3.3.6 Heat sensitive components.- The use of heat shunts or cooling devices shall be mandatory when hand soldering components that are subject to damage by heat.

3.3.7 Cleaning electronic assemblies.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed. The cleaning fluid shall have a gentle flow, at no time exceeding a pressure of 5 psi. The temperature of the cleaning solution shall not exceed 125°F. After cleaning, there shall be no residual flux, surface contamination or detrimental staining of the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

3.3.8 Drying.- The assembly may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than room temperature drying method is used, the assembly shall be dried in a circulating air oven the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psi. The air or nitrogen shall be filtered to meet the following requirements:

a. Permissible dust particle size not to exceed 6 microns
b. Permissible moisture content limited to 2.12 grains per cubic foot (dew point 32°F maximum)
c. Dust particles shall not exceed 1000 particles per cubic foot.

3.3.9 Cementing of components.- The cementing of components shall be limited to those specified on the applicable drawings. Prior to cementing with compound the board and components shall be cleaned and dried.

3.3.10 Wire termination.- When soldering wires to terminals or solder points, a small gap of 1/32 inch minimum to 1/16 inch maximum shall be left between the wire insulation and the solder build up. Burned, frayed or otherwise damaged wire insulation shall not be acceptable. Stripping of insulation shall be accomplished without dis-arranging, cutting or otherwise damaging any strand of the conductor. Heat shall not be used to remove outer jacket from shielded wire. Only mechanical methods of removal are permitted.

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3.4 Cleanliness of cavity assembly.- The cavity assembly shall meet the cleanliness requirements of detail drawings for component parts and assemblies. It shall be completely free of any foreign matter, clean, and dust free. The external window of the Cover Assembly 10540858 shall also meet this requirement.

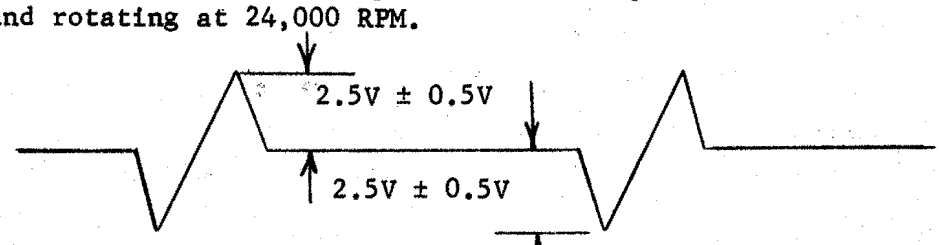
3.5 Vibration.- The cavity assembly mounted on its mounting flange shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the cavity shall show no evidence of physical damage and shall meet the requirements of 3.8 through 3.14.

3.6 Shock.- The cavity assembly mounted as in 3.5 shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half sine wave with a time duration of 11 ± milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the cavity assembly shall show no evidence of physical damage and shall meet the requirements of 3.8 through 3.14.

3.7 Storage temperature.- The cavity assembly shall withstand thermal stabilization at temperatures of -80° and +160°F. Subsequent to exposure to storage temperatures, the cavity assembly shall show no evidence of physical damage and shall meet the requirements of 3.8 through 3.14.

3.8 Sealing.- The cavity assembly shall be flushed with dry nitrogen until the gaseous mixture bled from the cavity has a dew point at least as low as -25°F. The interior of the cavity assembly shall be capable of maintaining 5 pounds per square inch gage pressure for a period of 2 hours at a temperature of 75° ± 2°F over which time the cavity leakage shall not exceed 0.2 psig. Subsequently, the pressure shall be reduced to normal atmospheric pressure, and the cavity sealed. The interior atmosphere of the accepted cavity assembly shall be nitrogen.

3.9 Magnetic Sensor output.- The output of the sensor measured from pin "B" to "A" of Connector 14J2 shall have the wave form as shown below when the motor is energized with pin "D" of 14J2 positive with respect to pin "E", and rotating at 24,000 RPM.



3.10 Temperature compensating network.- When the cavity assembly is thermally stabilized at the temperatures specified in column "A", the resistance value of the temperature compensating network measured between

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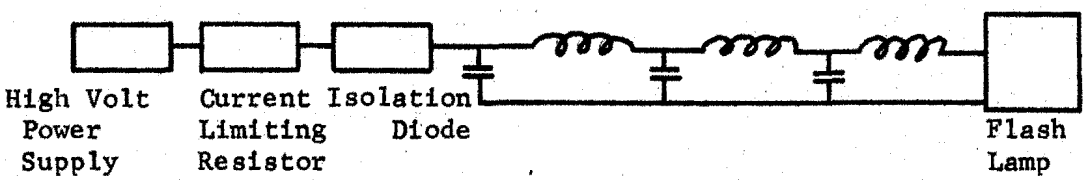
pins "C" and "F" of connector 14J2, shall be as specified in column "B".

A	B
75° ± 2°F	13,600 ± 250 ohms
-65° ± 2°F	18,100 ± 300 ohms
+155° ± 2°F	7,100 ± 250 ohms

3.11 Cavity assembly firing.

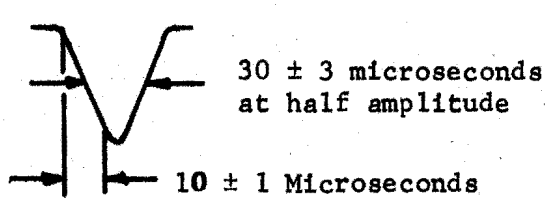
3.11.1 Operating conditions.- The cavity assembly shall meet the requirements of 3.11.2 to 3.11.4 inclusive while operating under the following conditions:

a. An operating voltage as specified in 3.11.2 Column "C" shall be applied to the flash lamp by a 250 microsecond, three section pulse forming network consisting of three 25 ± 0.1 microfarad capacitors and three 34.3 microhenry inductors as shown below:

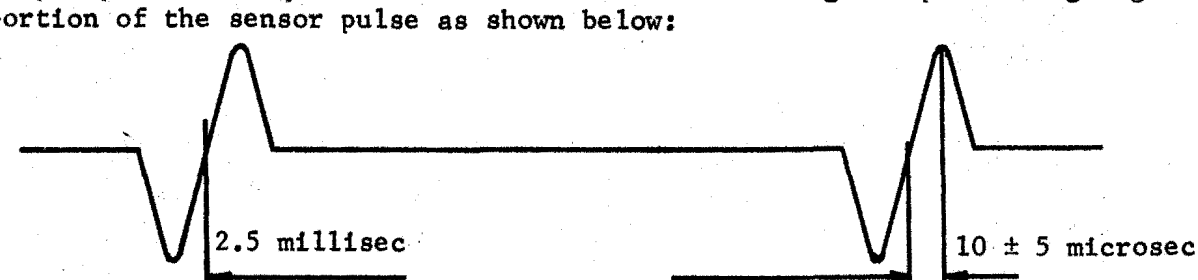


b. The cavity motor shall be energized with a D.C. voltage and adjusted until the "Q" switch prism attains a speed of 24,000 RPM.

c. The flash lamp shall be capable of triggering when a 13 ± 2 kilovolt negative pulse is applied, having the waveform as shown below:



Any overshoot or ringing in the trigger pulse shall not exceed 10KV in amplitude. The flash lamp shall trigger when the period of the sensor output pulses is equal to 2.5 milliseconds and during the positive going portion of the sensor pulse as shown below:



3.11.2 Power output.- When the cavity assembly is energized as specified in 3.11.1, a single laser pulse of peakpower output specified in

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column "B" shall be generated at the voltages specified in Column "C" while thermally stabilized at the temperatures indicated in Column "A".

A	B	C
+75° ± 2°F	3.0 megawatts ± 10%	*Vo (1650 volts max.)
-65° ± 2°F	1.0 megawatt minimum	Vo - 150V ± 50V
+155° ± 2°F	2.5 megawatts minimum	Vo + 250V + 50 - 75

* The operating voltage which has been established and marked on the cavity.

3.11.3 Pulse shape.- The rise time of the single pulse output meeting the requirements of 3.11.2 at +75° ± 2°F shall not exceed 15 nanoseconds when measured between the ten percent and ninety percent levels. The width of the pulse measured between the fifty percent peak power points shall not exceed fifty nanoseconds.

3.11.4 Output beam width.- At least seventy-five percent of the power output of a single laser pulse, meeting the requirements of 3.11.2 at 75° ± 2°F, shall be incident within the area of a circle having a diameter which subtends an arc of 5 milliradians in the far field.

3.12 Electrical continuity.- The continuity of the electrical circuits of the cavity assembly shall conform to wiring diagram Drawing C10540872 and schematic Drawing C10540881.

3.13 Insulation resistance.- The insulation resistance between each of the flash lamp contacts, the contact for the flash lamp trigger wire and the ground of the cavity assembly case shall not be less than 1000 megohms when subjected to a minimum potential of 500 volts direct current.

3.14 Interface requirements.- The cavity assembly shall interface with:

(a) Mounting seat, locating pin, pilot bore and mounting holes of Housing 10540511.

(b) Receptacle 10540690 (13P1) of telescope assembly.
(c) Plug 10540981 (11P3) of left housing assembly.
(d) Terminal 10541011 (23J1) of high voltage trigger.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65% Defective		
101. Electronic workmanship	3.3.1	Visual
102. Installation of hardware	3.3.2	SME; Visual
103. Spacing of parts	3.3.3	SME; Visual
104. Symbols and markings	3.3.4	Visual
105. Bend radius	3.3.5	SME; Visual
106. Heat sensitive components	3.3.6	Visual
107. Wire termination	3.3.10	SME; Visual
MINOR: NONE		

NOTE: The above characteristics shall be inspected at in-process intermediate inspection stations.

TABLE II - CLASSIFICATION OF DEFECTS

Use 100% inspection

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
108.	Sealing	3.8	4.6.2
109.	Magnetic sensor output	3.9	4.6.5
110.	Temperature compensating network (+75° ± 2°F)	3.10	4.6.6
111.	Temperature compensating network (-65° ± 2°F)	3.10	4.6.6
112.	Temperature compensating network (+155° ± 2°F)	3.10	4.6.6
113.	Power output (+75° ± 2°F)	3.11.2	4.6.7.1, 4.6.7
114.	Power output (-65° ± 2°F)	3.11.2	4.6.7.4, 4.6.7
115.	Power output (+155° ± 2°F)	3.11.2	4.6.7.5, 4.6.7
116.	Pulse shape	3.11.3	4.6.7.2, 4.6.7
117.	Output beam width	3.11.4	4.6.7.3, 4.6.7
118.	Electrical continuity	3.12	4.6.3
119.	Insulation resistance	3.13	4.6.4
120.	Interface requirements	3.14	4.6.8

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production,

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including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Cleaning	3.3.7, 3.4
Drying	3.3.8
Cementing of components	3.3.9

4.4 Special sampling.

4.4.1 General.- Three cavity assemblies shall be selected at random as a special sampling from each 50 produced. The samples shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable Dwgs & Specs - Visual
302.	Fabrication	3.2	Applicable Dwgs Visual
303.	General specifications	3.3	MIL-F-13926 - Visual

NOTE: The tests in Table III shall be conducted at a temperature between +70° and +80°F.

4.4.2 Environmental.- Three cavity assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall be subjected to the tests in Table IV and shall then meet the requirements and tests of Table II.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperature (-80° and +160°F)	3.7	4.6.1.1
305.	Shock (between +70° and +80°F)	3.6	4.6.1.2
306.	Vibration (between +70°and +80°F)	3.5	4.6.1.3

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Cavity firing, input characteristics.- The special testing equipment utilized to energize the cavity assembly shall be capable of providing the input characteristics specified in 3.11.1.

4.5.2.3.2 Cavity firing.

a. Output Power.- The output of the laser cavity shall be measured by a method referrable to NBS standards with an accuracy of .2 megawatts. The test set-up and equipment shown in Figs. 1 & 2 is acceptable, but is not mandatory. Use of a non-specular (matte finish) target board of high reflectance (at least 85%), as shown in Fig. 1 is recommended to attenuate the beam before incidence on the detector. Dimensions shown are approximate, for illustrative purposes only. The detector shall be calibrated by means of NBS standard(s) to measure the output power of the laser in the specific set-up employed. The method of power measurement, supported by appropriate calibration data, shall be Government approved prior to being utilized in the final acceptance inspection of the cavity assembly.

b. The laser pulse beam width shall be measured using a circular aperture punched out of a mask of black flock material (max. reflectance 2%) and positioned on the target board as in Fig. 1. The diameter of the aperture is to be of such a size as to subtend an angle of 5 milliradians at a target board distance from the laser cavity of at least 20 feet to insure that the measurement is being made in the far field. A holding device shall position and clamp the cavity assembly exit aperture relative to the center of the five milliradian target area. The holding device shall

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be pre-aligned to the target by utilizing a set-up master simulating the mounting flange and exit aperture of a cavity assembly as shown in Figure 3. With the set-up master assembled to the holding device, the exit aperture axis shall be aligned perpendicular to the center of the five milliradian target area (see Figure 1).

4.5.2.3.3 Sealing.- Special pressure testing equipment to test sealing of the cavity assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester Two-Station.

4.5.3 Standard test equipment.- In conjunction with 4.5.2.3, the following standard test equipment shall be utilized in performance of the applicable tests as specified in 4.6.

Nomenclature	Description
1. Oscilloscope (CRO with camera) See Figures 1 & 2	Frequency response, DC to 85 megacycles; Rise time, 4.5 nanoseconds; Input impedance 10,000 megohm or greater; Sensitivity 50 MV/CM; Sweep time .1 microsecond per centimeter, no parallax graticule; Camera lens F/1.9, 1:05 object to image ratio; Film ASA equivalent of 10,000 (Polaroid Type 410 or equivalent).
2. High voltage power supply (DC)	Range 0-2500 volts, current rating 200 milliamperes with 50,000 ohm, 20 watt limiting resistor.
3. High voltage digital voltmeter	Range 0-3000 volts minimum. Accuracy within 0.1% of reading or .01% of full scale. Input impedance 100 megohms greater.
4. Ammeter D.C.	Range 0 to 150 microamperes minimum. Accuracy within 5% at 150 microampere scale reading.
5. Low voltage power supply (DC)	Range 0-24 volts, current rating 2 amps.
6. Wheatstone bridge	Accuracy within .05% at required readings. See paragraph 4.6.6 for additional requirements.

4.6 Test methods and procedures.

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4.6.1 Environmental.- The cavity assembly shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the cavity assembly shall be subjected to the tests specified in 4.6.2 through 4.6.8.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.7.

4.6.1.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.7.

4.6.1.3 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The cavity assembly shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.5. The vibration test shall then be performed in accordance with Method 514 Procedure 1, Part 1, of MIL-STD-810.

4.6.2 Sealing.- Internal pressurizing of the cavity assembly, to test sealing, shall be performed with special testing equipment conforming to 4.5.2.3.3. A dew point tester shall be utilized with this test. The cavity assembly shall be flushed with dry nitrogen gas until the gaseous mixture bled from the cavity assembly has a dew point at least as low as -25°F, as specified in 3.8. Pressurize the cavity assembly with dry nitrogen gas to the value specified in 3.8, in accordance with the procedure for internal pressure testing of Drawing F8565556. Subsequent to reaching the pressure specified, the source of pressure to the cavity shall be shut off. At the end of the 2 hour period, the leakage shall not have exceeded 0.2 psig as indicated on the pressure gage of the testing equipment, to determine conformance with 3.8. Upon satisfactory completion of the sealing test, the internal pressure shall be reduced to the normal pressure specified in 3.8. The cavity assembly shall then be sealed.

4.6.3 Electrical-continuity.- The wiring of the cavity shall conform to the respective schematic drawings, utilizing standard measuring equipment and a visual inspection to determine conformance with the requirements of 3.12.

4.6.4 Insulation resistance.- Insulation resistance shall be measured between each of the isolated circuits and ground as indicated in 3.13. Measurement shall be made by means of a megohm bridge.

CAUTION: Do not apply megohm bridge used for this test to Connector 14J2.

4.6.5 Magnetic Sensor output.- The output of the sensor shall be tested utilizing an oscilloscope capable of measuring the wave form specified in 3.9 while the motor is rotating at 24,000 RPM.

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4.6.6 Temperature compensating network.- The following tests shall be conducted with the cavity assembly positioned in a test chamber that conforms to the "Test Facilities" requirements of MIL-F-13926 and conditions of 3.10. To measure the resistance value of the network across pins C and F of Connector 14J2, a wheatstone bridge shall be used. The maximum current through the network shall not exceed 150 microamperes. A current limiting resistance and an ammeter shall be used to control and read current. The resistance values of the resistance and the meter must be deducted from the value indicated by the bridge to obtain correct resistance readings of the network. With the network thermally stabilized for a period of 1 hour at each temperature specified in column "A" of 3.10 the resistance value shall be as specified in column "B" of 3.10.

4.6.7 Performance.- The following tests shall be conducted with the cavity assembly positioned in a temperature test chamber that conforms to the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.11.1 through 3.11.4. The test equipment specified in 4.5.2.3.1, 4.5.2.3.2, 4.5.2.3.3 and 4.5.3 utilized in these tests, shall be positioned external to the chamber with the necessary leads and connections made to the cavity assembly in the chamber. The test chamber shall be provided with an opening to permit firing of the cavity assembly as shown in Figure 1. The 5 milliradian black target shall be removed from the test set-up as shown in Figure 1, when performing the tests specified in 4.6.7.1, 4.6.7.2, 4.6.7.4 and 4.6.7.5.

NOTE: Caution must be exercised to avoid accidental contact with the high voltage terminals of cavity assembly or associated test circuits during test.

4.6.7.1 Power output (+75°F).- Perform this test after the cavity assembly has been thermally stabilized for a period of one hour at 75° ± 2°F. While still at the temperature specified, apply the V_o marked on the cavity assembly under test. The cavity assembly shall be fired at least 5 times at 5 minute intervals. The average obtained shall meet the requirements specified in Column B of 3.11.2. This test shall be repeated at the conclusion of the low and high temperature tests specified in 4.6.7.4 and 4.6.7.5.

4.6.7.2 Pulse shape.- Conduct this test in conjunction with the test specified in 4.6.7.1. The single pulse rise time shall meet the requirements specified in 3.11.3.

4.6.7.3 Output beam width.- Perform this test at the temperature established in 4.6.7.1. The black target shall cover the white target and the center of the 5 milliradian aperture shall be aligned to the exit aperture axis of the set-up master (see Figures 1 and 3). The cavity assembly shall be fired 3 times and the average peak power output computed. The average peak power output shall be greater than 75% of the average peak power determined in 4.6.7.1. The cavity assembly shall meet the requirements specified in 3.11.4. A period of 10 minutes shall elapse between each firing.

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4.6.7.4 Power output (65°F).- The cavity assembly shall be thermally stabilized at -65°F for a period of 1 hour. At the end of the 1 hour period, and while still at the temperature extreme, apply an operating voltage specified in Column C of 3.11.2 to the cavity assembly. The cavity assembly shall be fired a minimum of 5 times. A period of at least 10 minutes shall elapse between each firing. The cavity assembly shall meet the requirements of Column B of 3.11.2 upon application of an operating voltage within the range specified in Column C.

4.6.7.5 Power output (+155°F).- Perform this test as specified in 4.6.7.4, except that the cavity assembly shall be thermally stabilized at +155°F. The cavity assembly shall meet the requirements of Column B of 3.11.2 upon application of an operating voltage within the range specified in Column C.

4.6.8 Interface requirements.- Perform this test using 2 test housings. One housing designed to the maximum limit of each interface with a tolerance of 10% of total limit, and 1 designed to the minimum limit of each interface with a tolerance of 10% of total limit. Do not extend any interface beyond maximum or minimum limits. All pertinent interfaces shall be provided as depicted on Drawing F10540511 as specified in 3.14. The cavity shall be capable of being properly positioned and secured to both test housings. The cavity assembly shall interface with a test receptacle, a plug and terminal as specified in 3.14 designed with pertinent interfaces conforming to the requirements of drawings specified. Continuity between all connectors and test connectors shall be indicated when fully engaged.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11727549. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This cavity assembly is a component of the Laser Range Finder, AN/GVS-3 and is the unit within and from which the laser output pulse of the range finder is generated and released.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing (see 5.1).
c. Applicable packaging data sheet number (see 5.1).
d. Provisions for First Article testing.

Custodian:

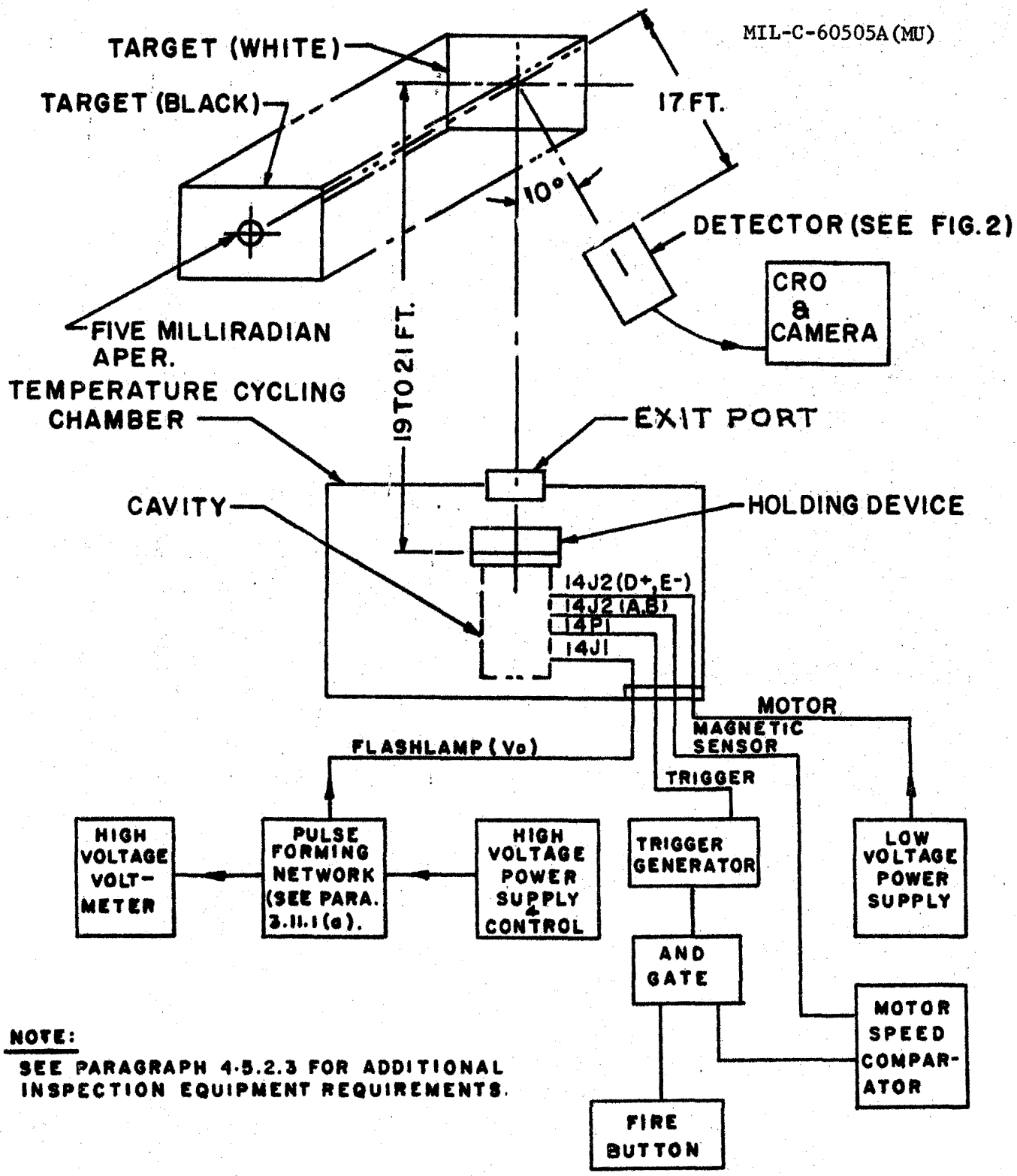
Army - MU

Preparing activity:

Army - MU

Project No. 1240-A170

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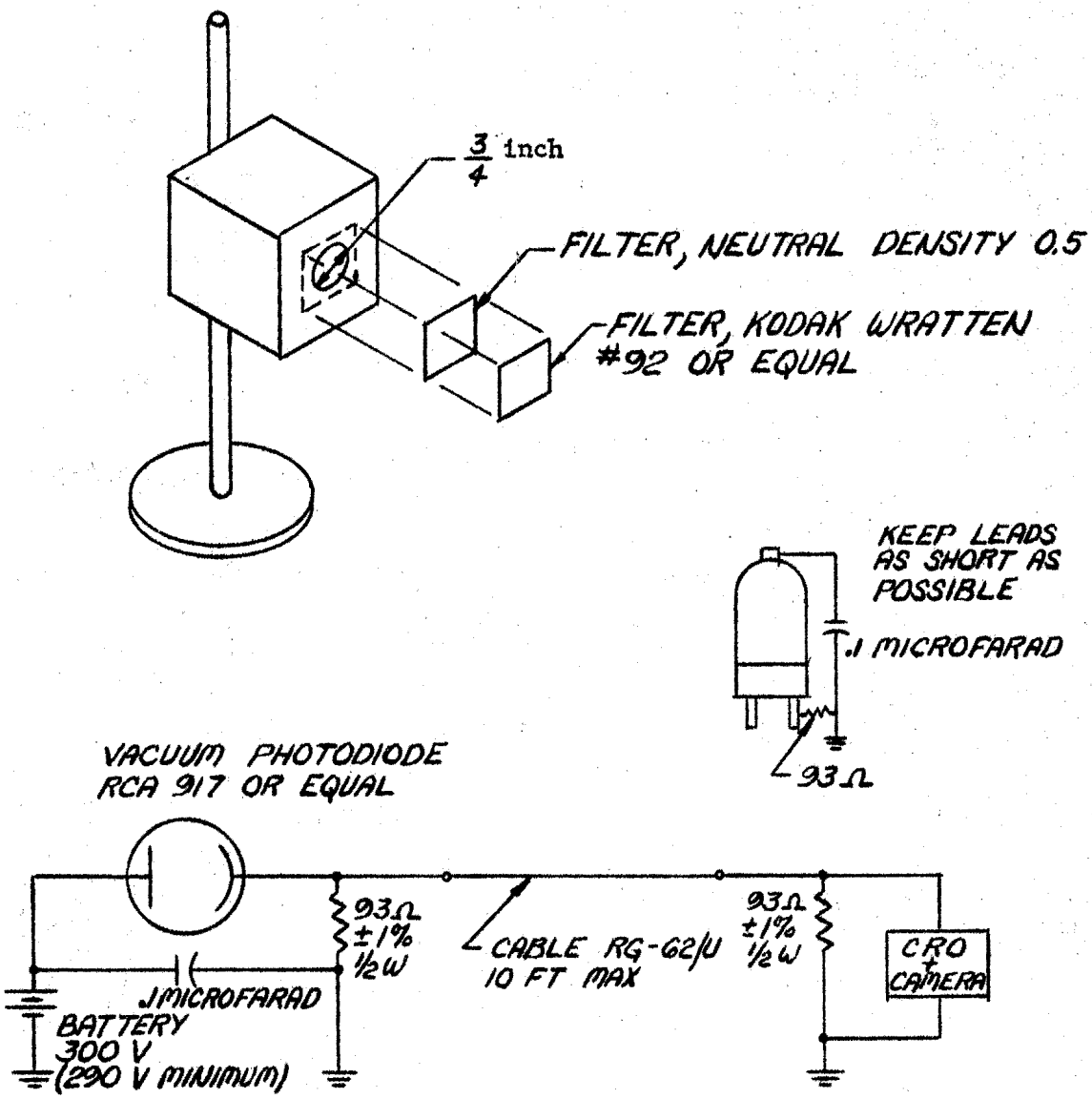
NOTE:

SEE PARAGRAPH 4.5.2.3 FOR ADDITIONAL INSPECTION EQUIPMENT REQUIREMENTS.

TYPICAL CAVITY FIRING TEST SET-UP

FIG. 1

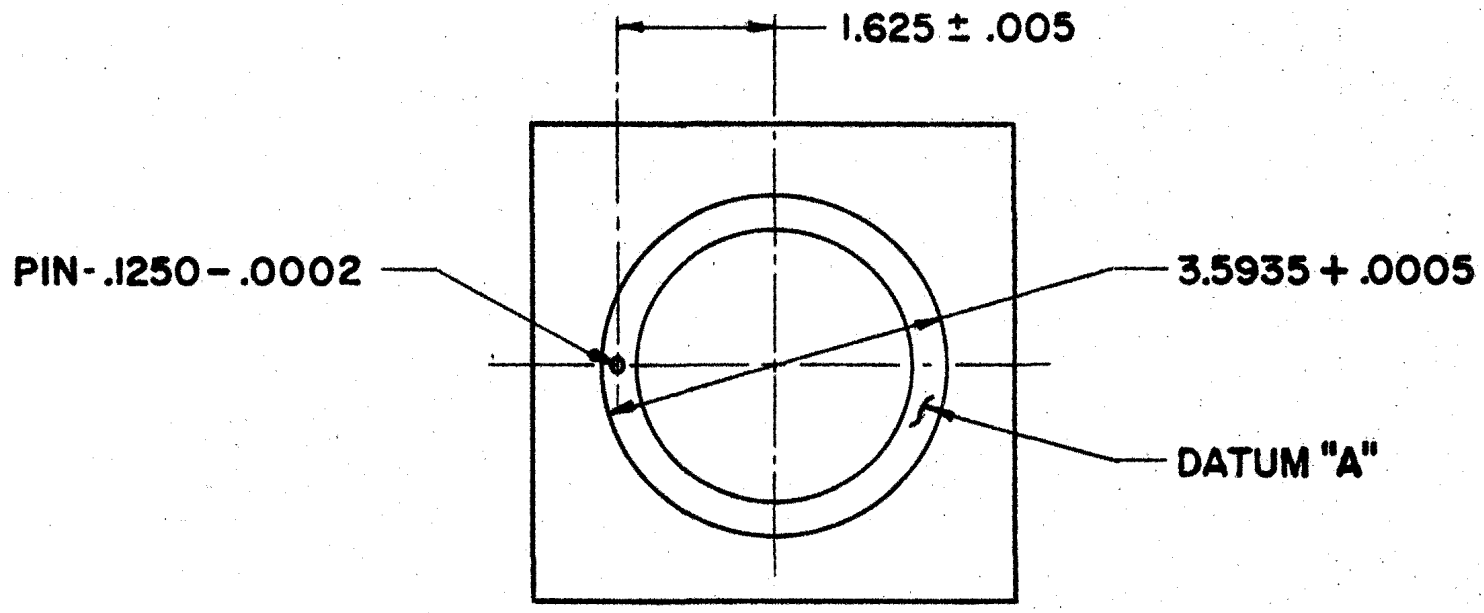
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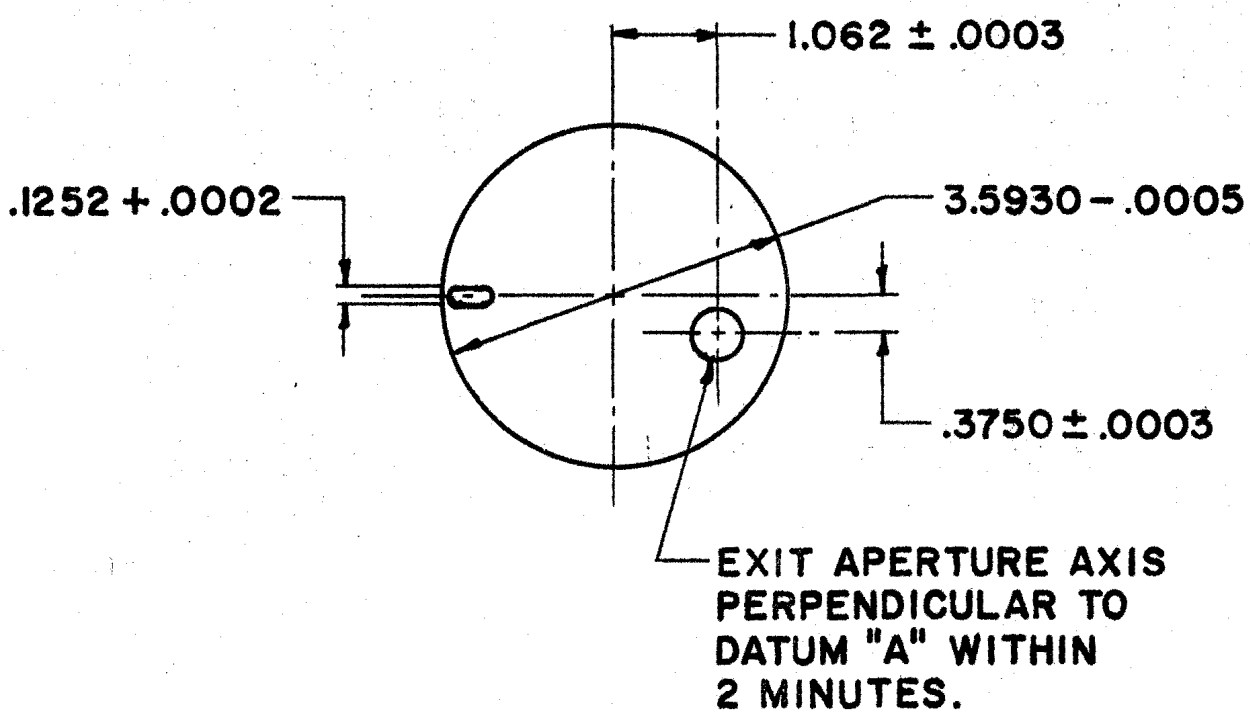
TYPICAL DETECTOR FOR POWER MEASUREMENTS

FIG. 2

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HOLDING DEVICE-ESSENTIAL FEATURES



SET-UP MASTER-ESSENTIAL FEATURES

FIG. 3

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